Automotive Electronics- Sensors & Actuators

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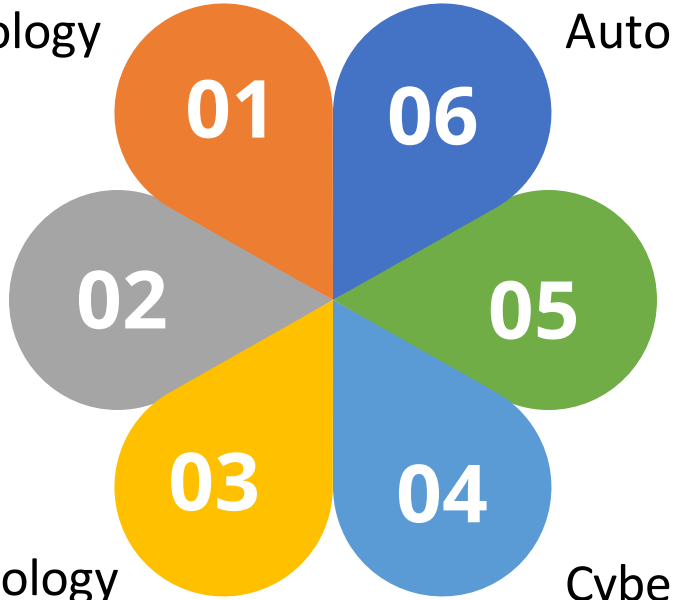
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Our Expertise

Electric Vehicle Technology

Future Mobility Overview

ADAS & Autonomous Technology



Automotive Electronics

IoT, Connectivity & Advanced Computing

Cyber Security for Automotive

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Introduction

- Sensors are now a vital part of any modern automobile design, serving many different purposes.
- They are instrumental in helping car manufacturers to bring models to market that are safer, more fuel efficient and more comfortable to drive.
- Over time, sensors will also enable greater degrees of vehicle automation, which the industry will benefit from.
- Computer controlled systems continually monitor the operating condition of todays vehicle.
- Through sensors ,computers receive vital information about a number of conditions, allowing minor adjustment to be made far more quickly & accurately than mechanical system.
- This session describes several specific sensors used in automobiles.

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What is a Sensor?

- A sensor is an electronic device that monitor, detects and responds to some type of input from the physical environment ,and
- It sends information to the driver or ECU (Electronic Control Unit).
- The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.



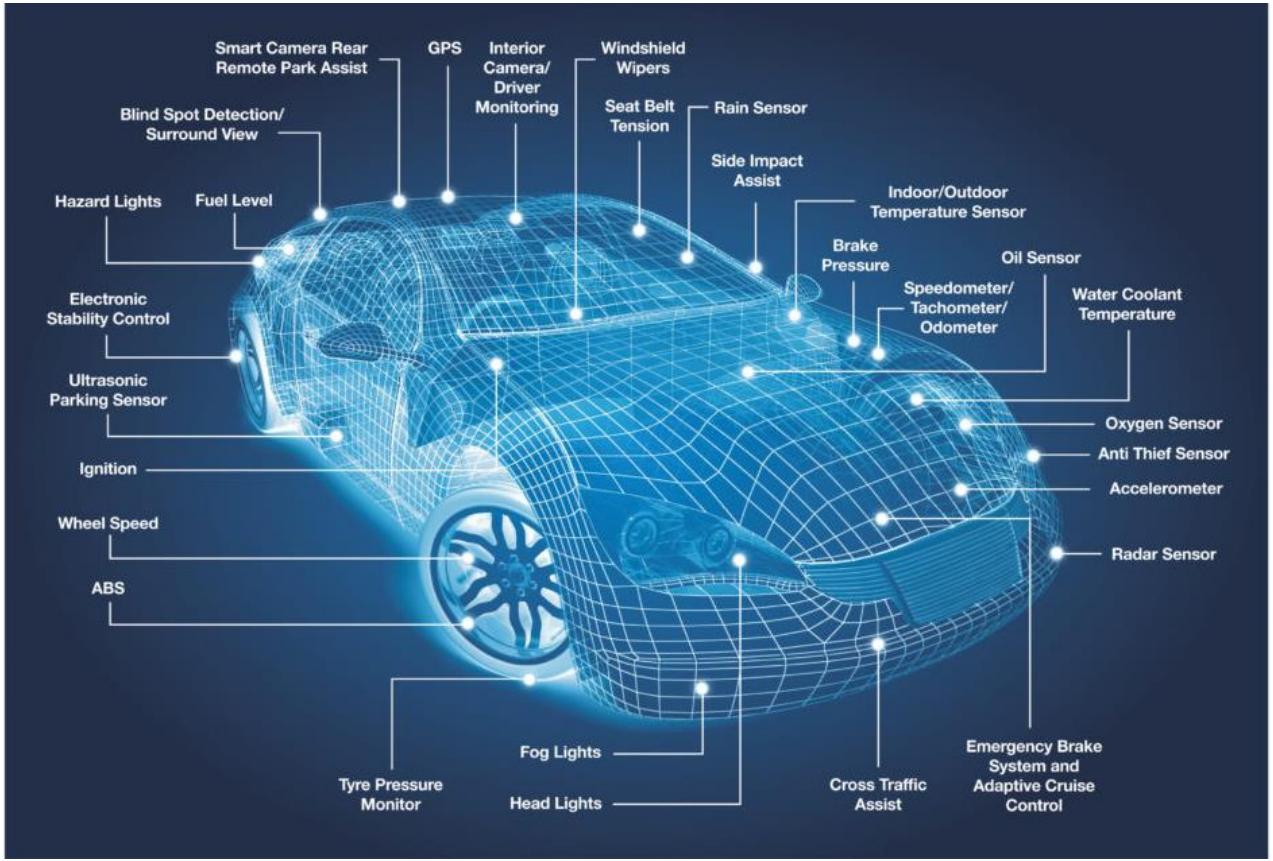
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Benefits of Sensors

- Sensors made life easy for drivers.
- Faulty components are easily detected
- Automatic control is common with vehicles that feature sensors.
- The engine is properly maintained with the devices.
- Every adjustment is done precisely with sensors.
- The driver receives warning information on bad components.

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Sensor Utility Details



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Sensor details

Sr. No.	Sensor	Main Function
1	The Mass Air Flow Sensor	Calculates the density and the volume of the air taken in by the engine
2	The Engine Speed Sensor	Monitors the spinning speed of the crankshaft
3	Oxygen Sensor	Measures the amount of unburden oxygen presented in the exhaust pipe
4	Manifold Absolute Pressure Sensor	Measures the manifold pressure inside and outside
5	Spark Knock Sensor	Ensures that the fuel is burned correctly
6	Fuel Temperature Sensor	Ensures the right amount of fuel is injected to keep the motion smoothly
7	Voltage Sensor	Manages the car speed and ensures the speed is controllable

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What are the types of sensors?

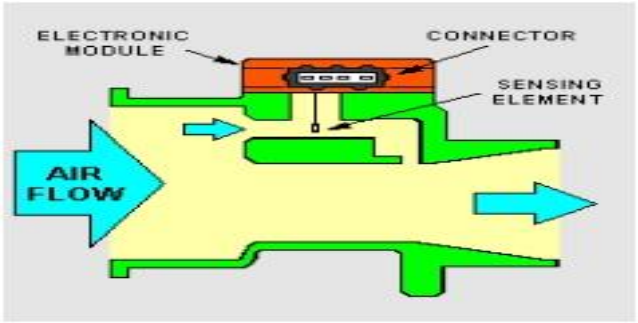
- Sensor can be categorized in multiple ways.
- Automobile sensors are intelligent sensors which can be used to control and process the pressure of oil, temperature, level of emission, coolant levels, etc.
- There are different types of sensors used in automobiles, but knowing the working of these sensors is essential.
- In order to the function of these sensors, here we have listed some popular sensors used in automobiles which include **the following slide…..**

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Mass Air Flow (MAF) Sensor

- **Principle of operation**: Operation of MAF is based on the principle of constant temperature, resistance-based sensor. Voltage supplied to balance the Wheatstone's bridge is directly proportional to the air flow.
- **Function :**A mass air flow (MAF) sensor responds to the amount of air flowing through a chamber containing the sensor.
  Air consumption is one of the basic parameters for calculating the necessary amount of fuel.
- This sensor can be controlled by a computer and can calculate the air density in the engine.
- **Note**- If the working of this sensor halts, then the running of the vehicle will be stopped.







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Inductive (VR) Sensor

- Inductive (magnetic reluctance) speed sensors are lightweight and very robust, which make them particularly suited to demanding motorsport applications.
- In an inductive sensor, the magnetic field around a permanent magnet changes if a ferrous target (e.g., a toothed wheel) is moved in front of the sensor.
- The change is sensed by the voltage generated in a coil of wire in the magnetic field.
- The coil and magnet are integrated in the sensor body for ease of installation.

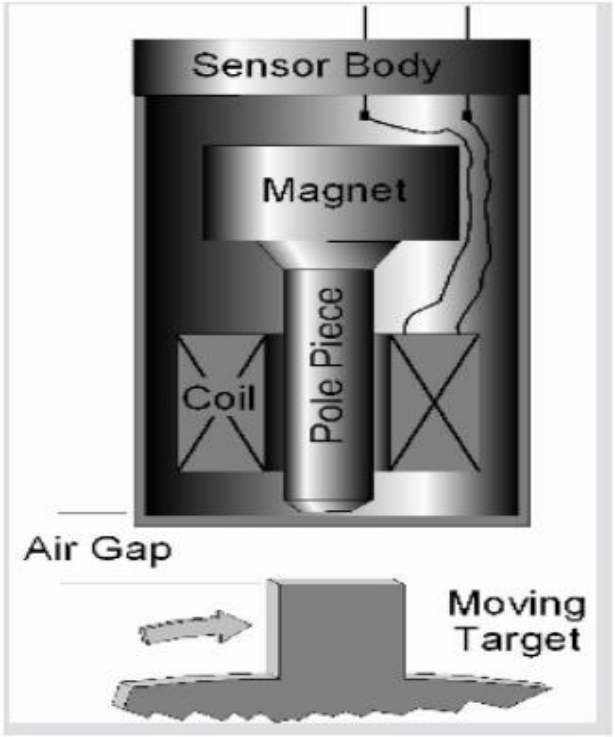
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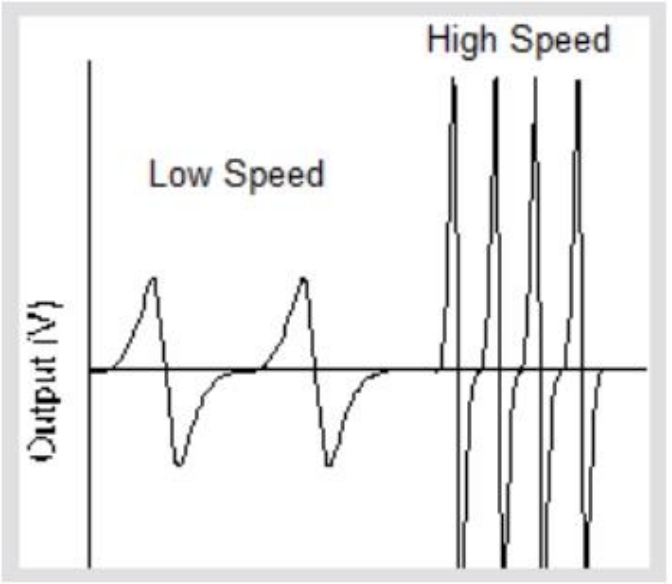
Inductive (VR) Sensor

- The magnitude of the induced voltage increases with the speed of movement of the ferrous object (i.e., the target wheel speed). The voltage decreases as the distance (air gap) between the end of the sensor and the moving target gets larger.
- Typical applications are measuring speed / position of crank and camshaft. Wheel speed and turbocharger speed.

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Inductive (VR) Sensor - Construction and output





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Oxygen (O_2) sensor





- **Principle of operation**: Discrete-type sensor or Self-generating DC Voltage
- **Function** : Located in the exhaust stream, usually near the exhaust manifold and after the catalytic converter, the oxygen sensor (or O2 sensor) monitors the content of exhaust gases for the proportion of oxygen.
- Measures air-fuel mixture from the exhaust and the catalytic converter's effectiveness (CAT).

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Oxygen (O_2) sensor



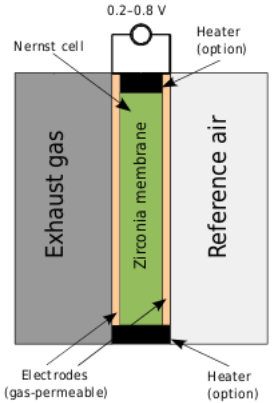


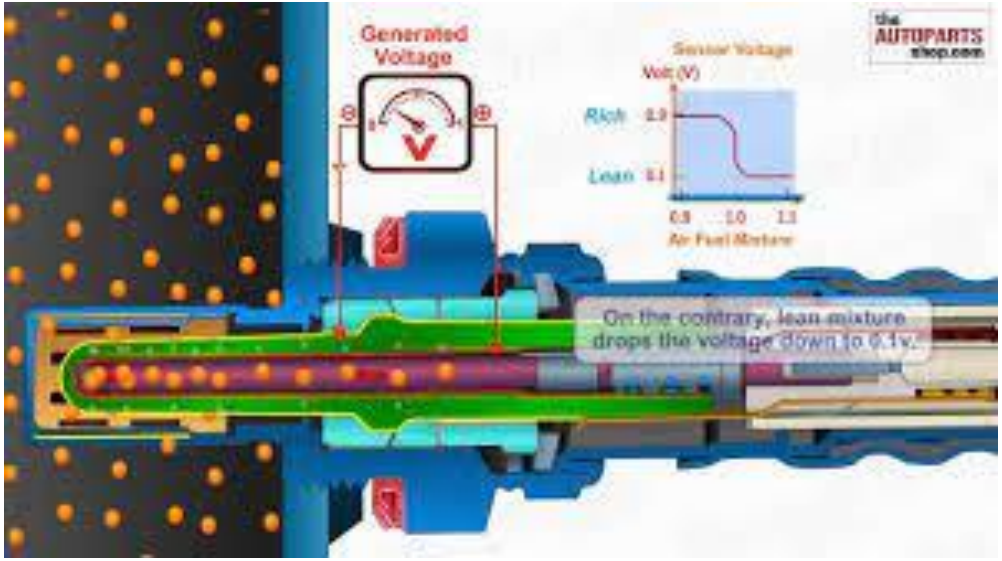
- Measures air-fuel mixture from the exhaust and the catalytic converter's effectiveness (CAT).
- The information is compared to the oxygen content of ambient air and is used to detect whether the engine is running a rich fuel ratio or a lean one. The engine computer uses this information to determine fuel metering strategy and emission controls.
- Two types—1.zirconium dioxide (ZrO2) – 2.titanium dioxide (TiO2).
- **Zirconium dioxide is the most commonly used.**

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Zirconia Oxygen Sensor

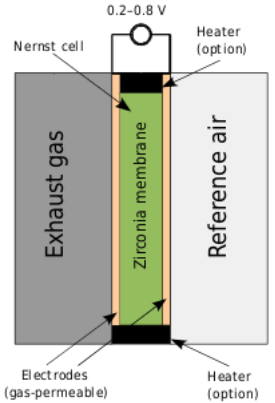
- Zirconia oxygen sensors use heat and chemistry to detect oxygen. Zirconia dioxide is coated with a thin layer of porous platinum to form a solid-state electrochemical fuel cell.



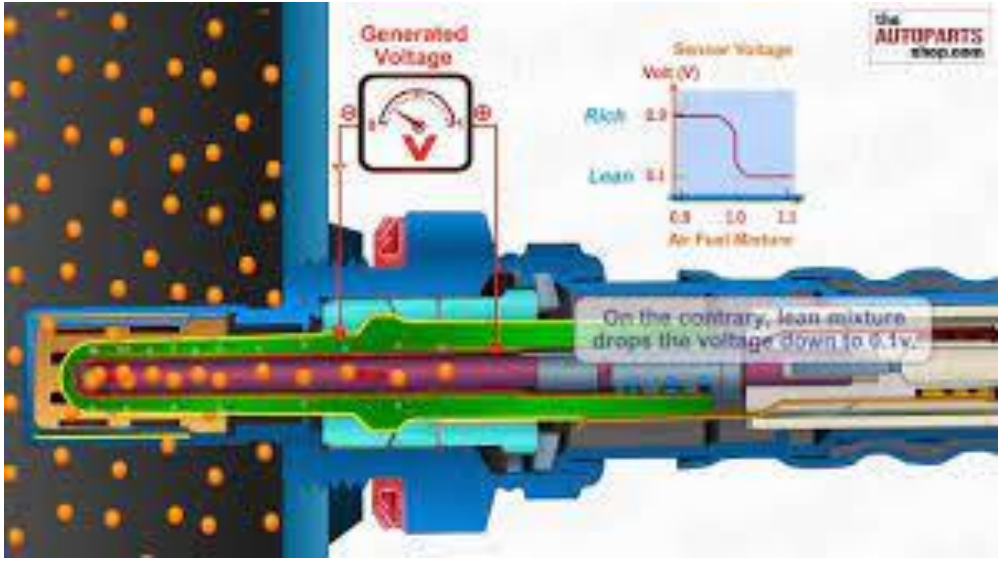


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Zirconia Oxygen Sensor



- Carbon monoxide, if present in the test gas, is oxidized by the oxygen to form CO2 which triggers a proportional flow of current.
- The zirconia sensor does not directly sense O2, but rather the difference between the concentration of oxygen in the sample gas and in fresh air.



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Coolant sensor



- **Principle of operation**: The RTD (Resistance Temperature Detector) is a temperature sensing device whose resistance changes with temperature. It provides coolant temperature values to EMS .
- **Function** : The coolant sensor is the most significant sensor used in automobiles. Because the computer depends on the sensor inputs to control all the functions.
- For instance, turn ON/OFF the EFE system (Early Fuel Evaporation), retard, spark advance, the flow of EGR, and canister purge.
- Generally, this sensor can be connected on the board. If the sensor is failed, then there will be some indications stalling, like poor fuel mileage, etc. So, the status of the sensor should be checked whether it is defective or not. If it is damaged, then it will be a problem.

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Thermistor principle

- Thermistors are variable resistors whose resistance changes in relation to temperature. Thermistors can have either a negative temperature coefficient **(NTC)** or a positive temperature coefficient**(PTC).**
- A thermistor with a negative temperature coefficient will decrease the resistance as the temperature is increased. On the other hand, a thermistor with a positive temperature coefficient will increase in resistance as the temperature is increased.
- The thermistor have two terminals one for power and one for ground. A reference voltage is supplied to one terminal through a fixed series resistor located inside the computer. The other terminal of the thermistor is connected to the ground usually back through the computer.

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Manifold Absolute Pressure(MAP)sensor





- **Principle of operation**: Piezo resistance i.e., Signal varies with pressure
- **Function** : The main function of this sensor in an automobile is to monitor the load of an engine.
- Mostly, it measures the dissimilarity among manifold pressure.
- This can be received from the outside pressure by the car to make sure that the car engine is capable to receive petroleum depending on the changes within the pressure.

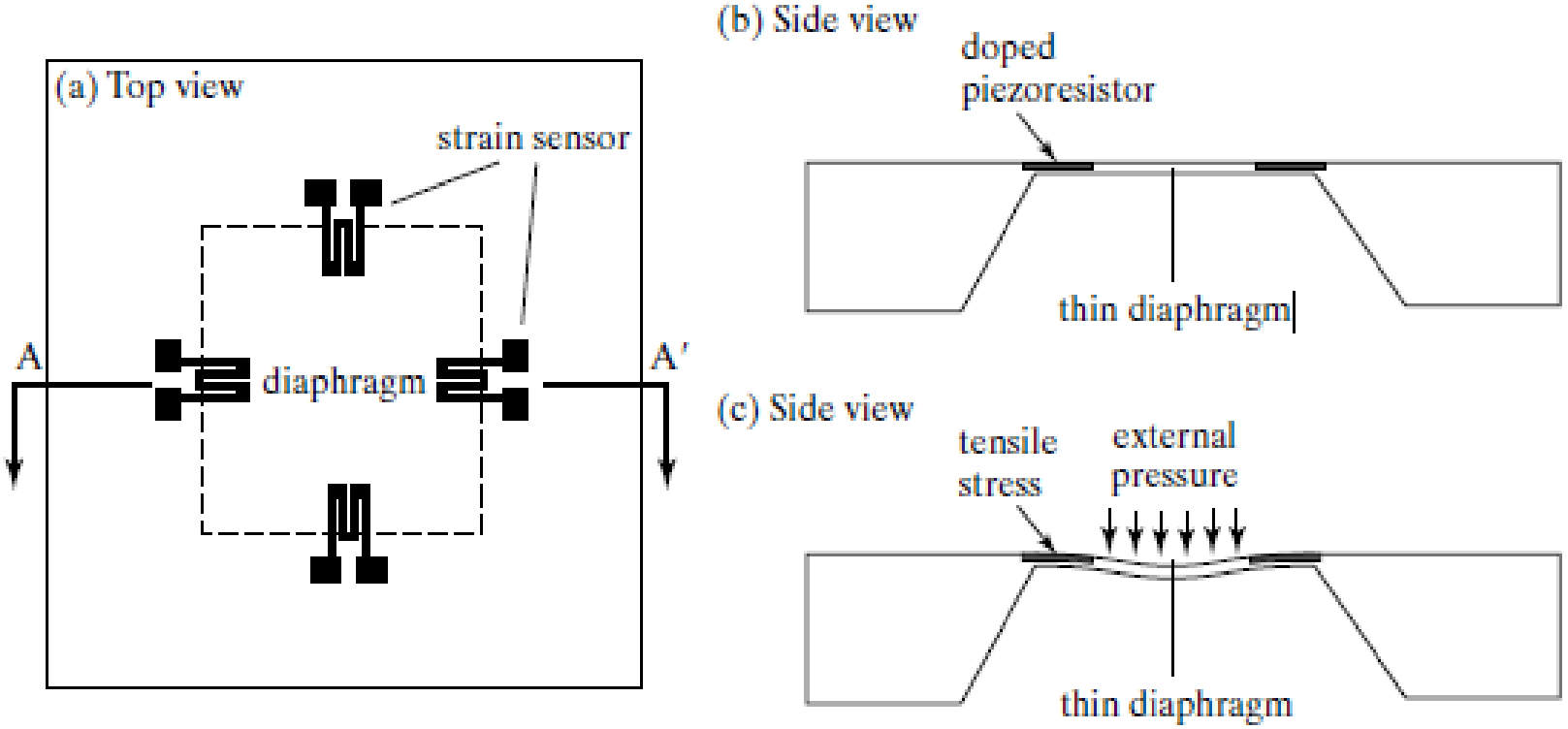
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Piezo Resistive Principle

- A semiconductor may resistor may change its resistance when applied with stress or deformation.
- This effect provides an easy and direct energy/signal transduction in mechanical and electrical domains.
- **Piezoresistive** refers to the change in **resistivity** due to stress
- If there is change in **resistance** due to stress or deformation, then that's a **strain gauge**. Stress is the cause and strain is the effect.
- The piezo resistors on the sensing element are four and are connected in a wheat stone bridge circuit.
- It's used in MEMS field for wide variety of sensing needs e.g., pressure, acceleration, flow, gyro rate etc.
- Bulk micro fabricated pressure sensors with thin deformable diaphragms made of single crystal .Silicon are the earliest products and still dominate the market today

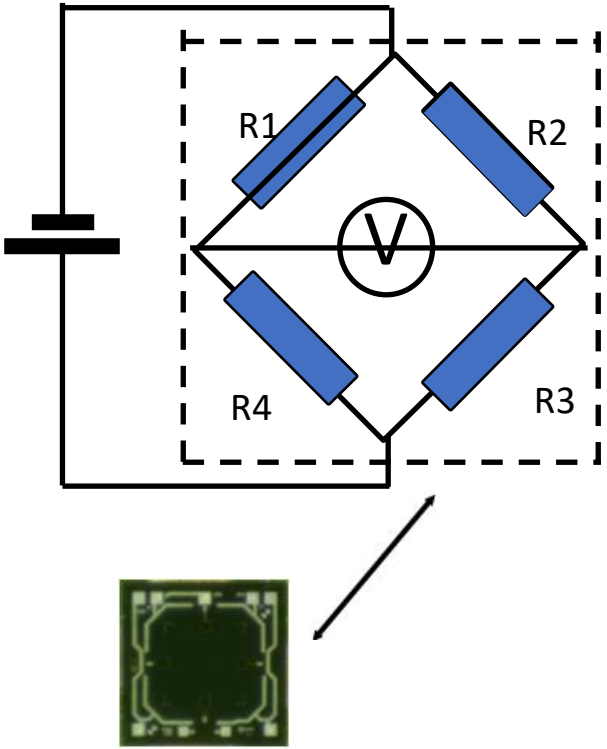
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Piezo resistive principle Diagram



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Wheatstone Bridge



The sensor has four implanted resistors in either an open or closed *configuration.*

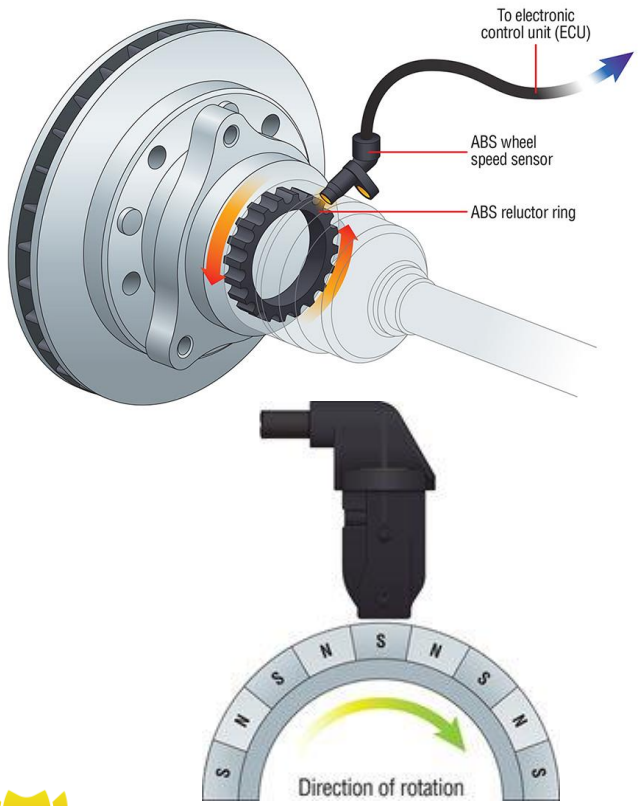
$$V1 = \frac{Vin.R4}{(R1+R4)} \qquad V2 = \frac{Vin.R3}{(R2+R3)}$$

$$Vo = V1\ V2$$

$$Vo = \frac{Vin.R4}{(R4+R1)} \quad \frac{Vin.R3}{(R3+R2)}$$

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Wheel speed or ABS sensor



- **Principle of operation**: Uses the Hall effect, which is a generation of a voltage (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and a magnetic field perpendicular to the current
- **Function** : An anti-lock braking system (ABS) is a vehicle safety system that allows the wheels of a car to maintain tractive contact with the road surface while braking, preventing the wheels from locking up (ceasing rotation) and avoiding uncontrolled skidding.

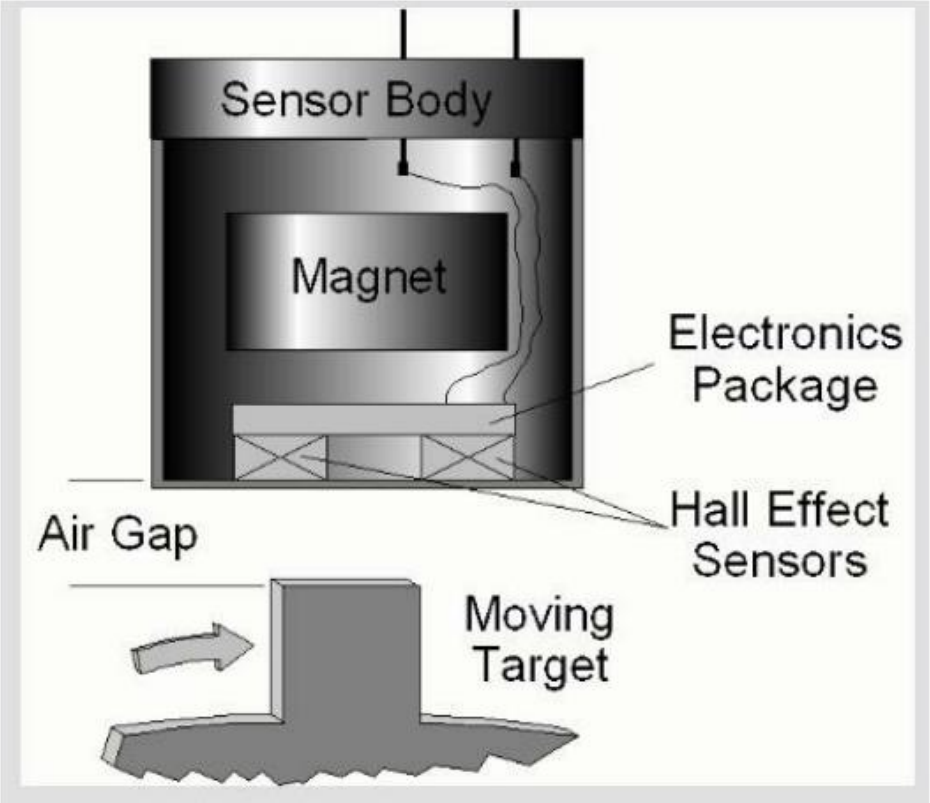
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Hall effect principle

- Using semiconductors (such as silicon), **Hall effect sensors work** by measuring the changing voltage when the device is placed in a magnetic field.
- In other words, once a **Hall effect sensor** detects that it is now in a magnetic field, it is able to sense the position of objects.
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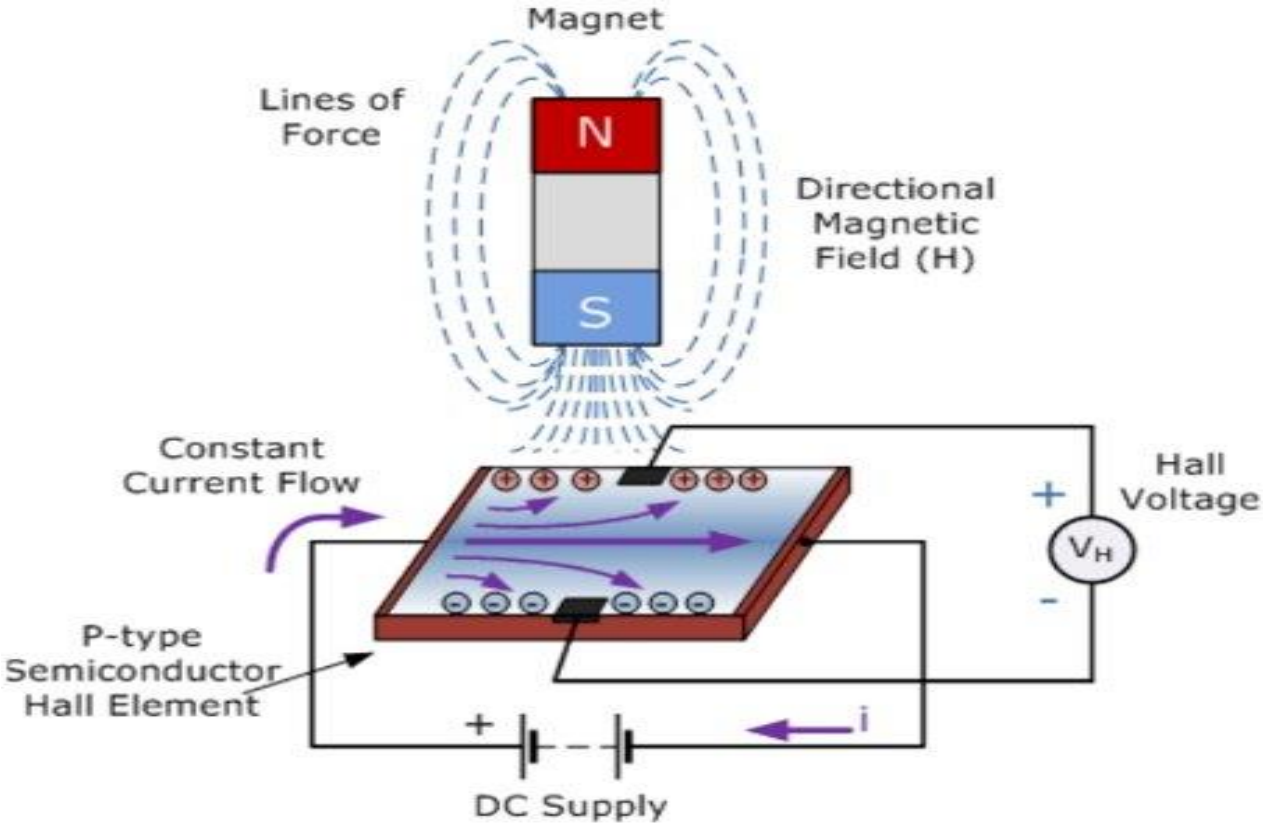
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Hall Effect how it works?



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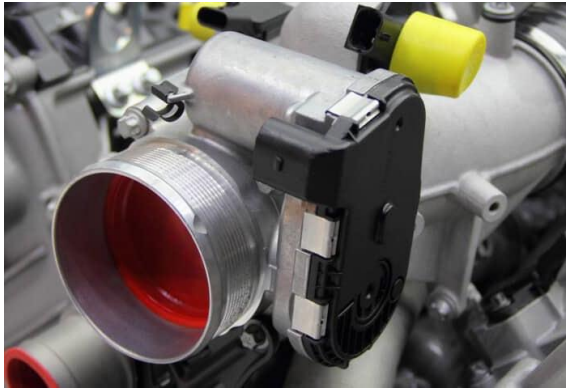
Hall Effect Principle- Diagram



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Throttle position sensor



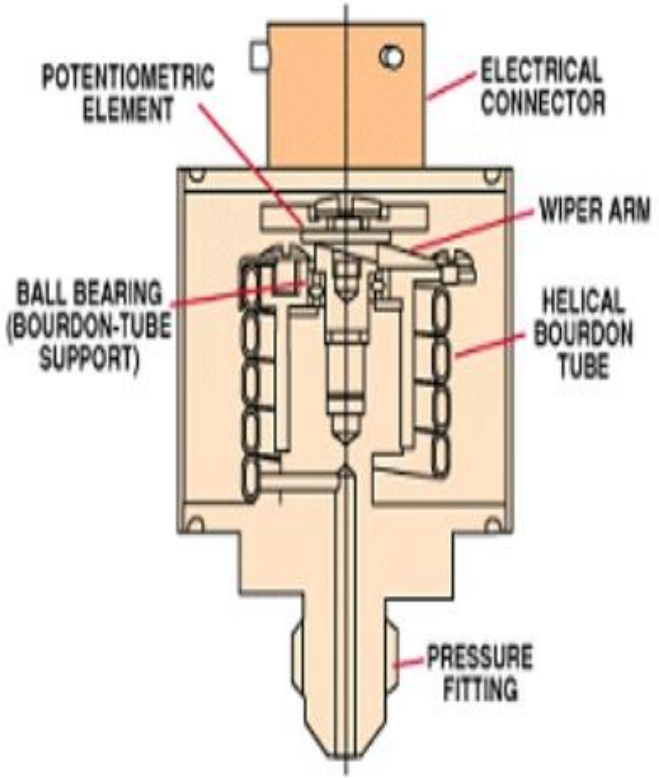


- **Principle of operation**: Based on Piezo resistance or potentiometer principle.
- **Function** : The throttle position sensor in automobile mainly uses feedback carburetion & electronic fuel injection (EFI). It informs the computer regarding the throttle opening rate as well as the position of the relative throttle.
- This sensor is a variable resistor, which is used to change the resistance as the throttle opens.
- **Note-** It is not complex to identify the faulty throttle position sensor symptoms. As there is a fall while speeding up, then you can identify the faulty of the sensor.

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Potentiometer Principle

- These use a Bourdon tube, capsule, or bellows to drive a wiper arm on a resistive element.
- For reliable operation, the wiper must bear on the element with some force, which leads to repeatability and hysteresis errors.
- These devices are very low cost and are used in low-performance applications.



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Rail/Fuel Pressure sensor

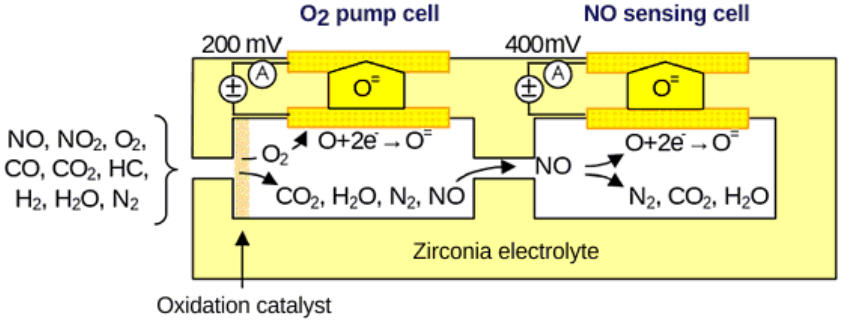


- **Principle of operation**: Pressure measurement results from the displacement of a steel diaphragm on which strain-gauge elements are located. These are connected in the form of a Wheatstone bridge. Voltage required to balance the bridge will be directly proportional to the pressure.
- **Function** : The fuel pressure sensor measures the fuel pressure on fuel pressure line or at the fuel pressure rail. It is most likely to be mounted on your fuel pressure rail, but sometimes it can also be mounted on the fuel pressure line.
- It is critical for the engine control module to measure the fuel pressure, as an increase in pressure would result in a richer fuel mixture, and a lower pressure would result in a leaner fuel mixture..

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Nox Sensor





- **Principle of operation**: Based on electrochemical principle. The sensor uses two or three electrochemical cells in adjacent chambers. The first cell electrochemically pumps O2 out of the sample so it does not interfere with the NOx measurement in the second cell. The need to remove O2 allows this type of NOx sensor to serve a dual purpose; it can also detect exhaust O2 level.
- **Function** : The NOx sensor is part of the NOx reduction aftertreatment system used in diesel vehicles with urea based SCR systems.
- The sensor located upstream of the SCR catalyst directly measures the engine-out NOx gas concentration, which helps determine the optimum amount of urea injection.

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Accelerator Pedal Sensor



- **Principle of operation**: Potentiometer is used in the accelerator pedal to transfer the signal which are nothing but the drivers input, processed at EMS ECU for handling the load through the air and fuel management
- **Function** : Accelerator pedal module converts driver's demand into signal which is fed to EMS ECU. It is equipped with set of 2 potentiometers which convert the pedal travel into voltage. It is mounted in driver's cabin.

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Sensors in EVs

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EV Battery current sensor

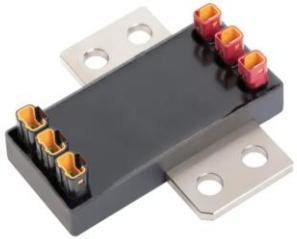
- EV current sensors are the basic components, used in the EV. They perform two major tasks.
  - They help us to know how much energy we use.
  - Also, the second task is avoiding overcurrent.
- Battery systems provide DC current. So, the battery current sensors perform DC current measurements. There are two main types of battery current sensors.
- EV current sensors can include resistive or magnetic elements based on their structure.

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Types of Battery current sensor

1.Shunt sensors: Shunt sensor-based current measurement devices use resistive elements. Sensor is usually non-isolated -HEVs/EVs

- Shunt sensors are robust and highly accurate. They have a simple structure. So, they are rather robust against thermal, mechanical, and electrical overloads. If we design the signal conditioning circuit well, the shunt sensors are perfect current sensors for automotive battery systems.



2.Magnetic sensor: Battery current sensor uses electromagnetic elements and measures the magnetic field of current.

- Therefore, by their nature, these types of current sensors are isolated. However, these sensors measure the current indirectly. Because of this, the accuracy could be less than the shunt sensors.



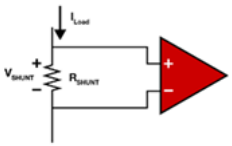
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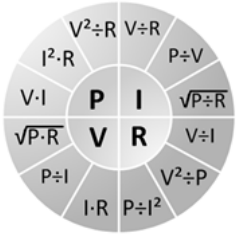
Ways of Battery current sensor

The physics behind Current Sensing

Ohm's Law (Shunt)

A current sense device directly measures the current through a relatively small ohmic valued (shunt) resistor.



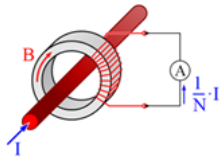


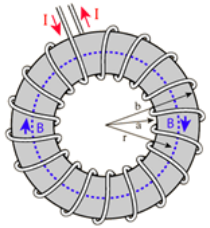
Attributes to consider:

- Common Mode Voltage Range & Isolation
- Accuracy & Calibration Needs
- Power Consumption & Dissipation
- Functionality & Space Constraints

Ampere & Faraday's Laws (Magnetic)

A current sensing device indirectly measures the current induced by a magnetic filed. Due to the non-invasive measurement technique, this method offers high levels of inherent isolation.



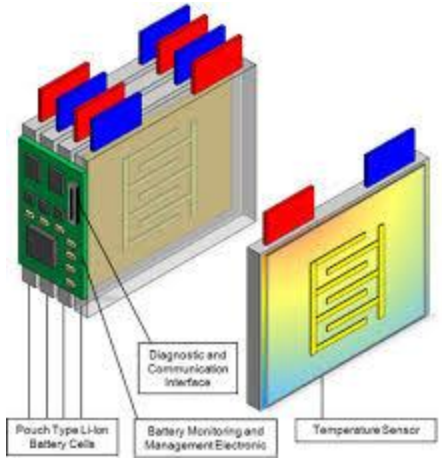


Attributes to consider:

- Isolation
- Dynamic Range
- Accuracy, Linearity and Sensitivity
- Integration & Size

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EV Temperature sensor



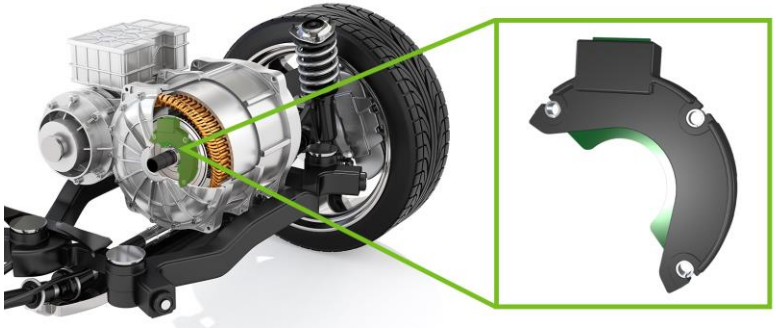


- **Principle of operation:** Based on NTC principle .
- NTC thermistor temperature sensor is a key component. These sensors provide essential temperature data for the battery to work in optimum condition, thus safeguarding the unit from potential hazards.
- Function :
- Temperature detection of electric motor
- Over-temperature protection of on-board charger
- Over-temperature protection of charging cable
- Over-temperature protection of charging pile
- Temperature detection of battery pack



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EV Motor Speed/Position Sensor



- **Principle of operation**: Based on Hall effect, resolver ,encoder type principle .

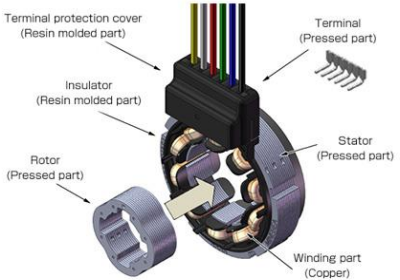
Function :

- To control and protect most valuable asset, the EV motor.
- Accurate motor sensing is key to building the future of EV, and high-speed position sensors allow us to collect information on vehicle propulsion and speed to understand how well the motor is performing.

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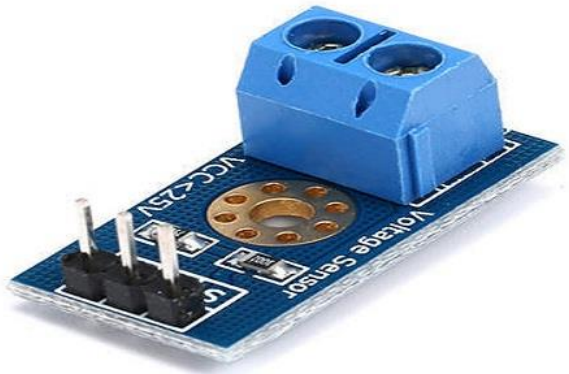
Resolver Type Motor Speed Sensor

- A resolver is a type of rotation angle sensor for controlling the drive motor that is the power source of EVs.
- It is used to detect the magnetic pole position of the motor and accurately grasp its rotational speed.



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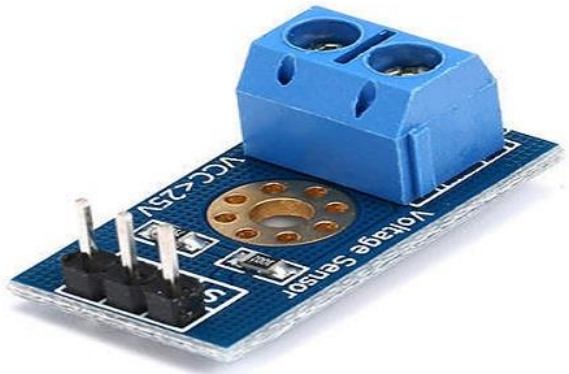
Voltage Sensor



- **Principle of operation**: Based on resistive or capacitive.
- The input of this sensor is the voltage, whereas the output is the switches, analog voltage signal, a current signal, or an audible signal.
- Some sensors provide sine waveforms or pulse waveforms like output & others can generate outputs like AM (Amplitude Modulation), PWM (Pulse Width Modulation) or FM (Frequency Modulation).
- The measurement of these sensors can depend on the voltage divider.

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Voltage Sensor

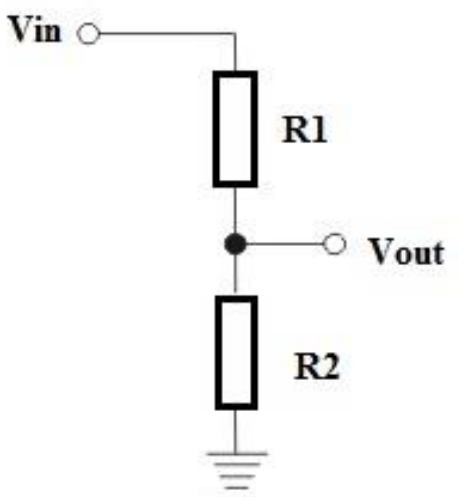


Functions:

- A voltage sensor is a sensor used to calculate and monitor the amount of voltage in an object.
- Voltage sensors can determine the AC voltage or DC voltage level.
- Voltage sensor which can be used to detect the range of voltage in any device. It decides the electrical charge within any device.

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Types of Voltage Sensor

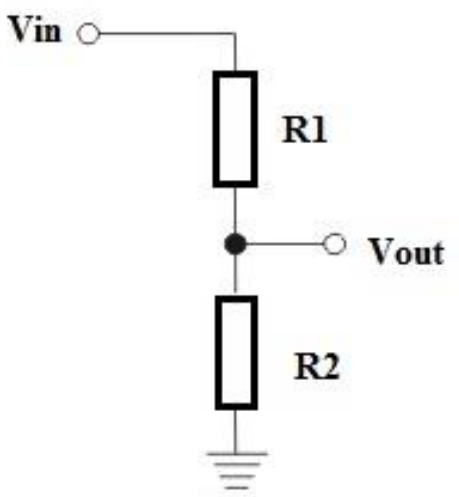


Resistor type sensors:
This sensor mainly includes two circuits like a voltage divider & bridge circuit.

- The resistor in the circuit works as a sensing element. The voltage can be separated into two resistors like a reference voltage & variable resistor to make a circuit of the voltage divider.

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Types of Voltage Sensor



- A voltage supply is applied to this circuit. The output voltage can be decided by the resistance used in the circuit. So the voltage change can be amplified.
- The bridge circuit can be designed with four resistors. One of these resistors can be subjected to the voltage detector device. The change in voltage can be directly exhibited. This difference alone can be amplified but the difference within the voltage divider circuit not only amplified.

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Actuators

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What is an Actuator?

- Actuators are mechanical or electro-mechanical devices that, upon being operated electrically, manually, or by various fluids, allow controlled and sometimes limited movements or positioning.
- They are devices which transform an input signal (mainly an electrical signal) into motion.
- Actuators basically need a control signal and a source of energy. Upon receiving a control signal, the actuator uses energy from the source to bring about a mechanical motion.

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Actuators in Automotive

- **Pneumatic:** they generate a force or resistance through air pressure. We could find them in the systems of tailgate lifting.
- **Hydraulic:** they are actuators that generate the movement from the displacement of fluids (valves of a variable suspension system).
- **Electromagnetic:** those based on electromagnetism, either through a magnet or an electromagnet. These are usually used in those mechanisms of the vehicle operating with electric current.
- **Gear motors or electric:** drives where a motor intervenes: brushless, broom, step by step…

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Actuator Utility Details

Actuators	Function
Fuel Injector	To inject fuel in to the cylinder as per requirement
Inlet metering valve	Used for metering the fuel at the inlet to the high pressure pump
Glow plugs	Cold starting aid, used for heating the combustion chamber
Electrically controlled EGR module	To control the EGR flow
Rotary Electric actuator Turbocharger	To control the air flow
Fuel Pump	To give the engine the fuel at the right pressure suitable for its work
Fuel pressure regulator	To control fuel injection rate
Idle speed actuator	To control air intake rate into engine
Spark Plug	To ignite fuel
Ignition coil	Energy storage device & transformer

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List of Actuators-Engine

- Fuel System
  1. Fuel Injectors(Electromagnetic and Piezoelectric actuators)
  2. Fuel Pressure Limiting Valve (Electromagnetic actuator)
  3. Throttle Body Actuator(Electromagnetic actuator)
  4. Electronic Choke Actuator (Heating actuator/Electromagnetic actuator)
  5. Fuel cut-off valve (Electromagnetic)
- Intake System
  1. Intake air throttle
  2. Variable Geometry Turbochargers (VGT) actuator
- Exhaust system
  1. Exhaust gas recirculation injector, pump motor, metering valve
  2. Solenoid valve, metering valve.

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List of Actuators-other system

- **Actuators for Climate Control System**
  1) Clutch for compressor engagement control
  2) Coolant/Refrigerant valve actuators
- **Vehicle System**
  1) Grill shutter actuator - For actuating the radiation grills to reduce the aerodynamics losses.
  2) Electric power sterling
  3) Electric coolant fan
  4) Bus door actuator for passenger safety
  5) Power windows

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List of Actuators-other system

- **Vehicle System**

  1) Central locking system actuators
  2) Headlight tilt actuator
  3) Windscreen/headlight wipers and washers
  4) Seat adjustment
  5) Sunroof adjustment

- **Aftertreatment system actuators**

  1) DEF injector
  2) Metering valves

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Actuator- Governing principles

Electromagnetic:

1. **Faraday's law of electromagnetic induction:** The voltage induced in a coil is proportional to the number of turns and rate of change of flux. The induced current flows in the direction that opposes the changing flux. Flux has no source or sink (What goes in comes out).
2. **Ampere's Law:** The magneto-motive force (mmf) around a closed loop is equal to net current enclosed by the loop. The objective of solenoid design is to transfer the maximum amount of NI (energy) from the coil to the working air gap.
3. **Flemings Left Hand Rule:**For direction of force (thumb), magnetic field and current.

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Non-Actuating Loads /Heating Actuators

- The actuators are major power consumers in any embedded automotive electronics.
- The non-actuating loads which don't really provide any physical motion or displacement but provide other type of physical quantity such as heat.
- The heaters, glow-plugs, defrosters etc. are said to be non-actuating loads.

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Non-Actuating Loads /Heating Actuators

- Since these also consume considerable power and have similar ways to control the power delivery at output with respect to control system; they are also considered as actuators in general.
- The limiting factor of operating temperature of a solenoid is the insulation material of the magnet wire used (Insulation classes).

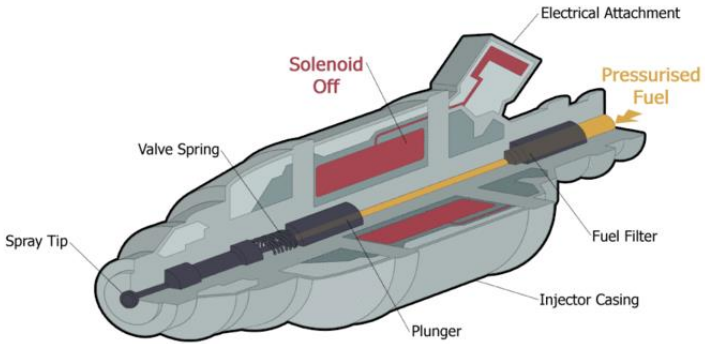
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Requirements for Actuators in Automobiles

- Best reliability for use in automotive environment
- Best efficiency in less size. (Power density)
- Easy construction, servicing, driving.
- Fit for environment
    1. Robust: (Withstands Vibrations experienced in automobile)
    2. Temperature: (Exhaust gas and Engine compartment temperature)
    3. Dirt ingress protection
    4. Oil and fuel protection

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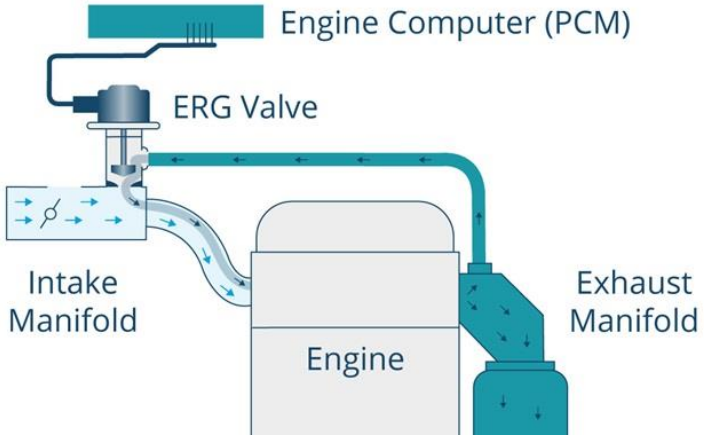
Fuel Injector



- **Principle of operation**: Injector solenoids are electrically actuated when the fuel injection in the cylinder is intended. Solenoid valve will lift the up and allows fuel in the fuel gallery in the injector. The needle at the nozzle end will be pressure lifted by the supplied fuel and the duration of the injection can be controlled through the solenoid actuation duration.
- **Function** : The purpose of the fuel injection system is to deliver fuel into the engine cylinders, while precisely **controlling** the injection timing, fuel atomization, and other parameters. The main types of injection systems **include** pump-line-nozzle, unit injector, and common rail.

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EGR Valve





- **Principle of operation**: Exhaust Gas Recirculation refers to the redirection of the exhaust gases after the combustion process and use it to achieve various goals in the engine of the vehicle.
- **Functions :** It helps reduce the amount of the nitrogen oxides (NOx) in the exhaust gases. Nitrogen oxides are normally formed in the process of combustion in the engine cylinders.
- However their formation increases dramatically at higher combustion temperatures (above 1600 °C or 2912 °F)
- The EGR system works by returning a small portion of exhaust gas to the engine's combustion chambers through the intake manifold, lowering combustion temperatures and therefore reducing the amount of NOx emitted.

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Throttle Body Actuator

- The throttle body actuator is an important part for engines operating on gaseous fuel.
- These are meant to control the air or air fuel mixture for these engines. It must regulate the air or mixture supply for the combustion engine
- The "throttle position feedback" sensor which is essentially a potentiometer can be a integral part of this actuator.
- This sensor provides the current angle of opening (degree) in terms of resistance (R).
- The actuators can be motorized or electromechanical solenoid type.
- Application: Engine speed governor

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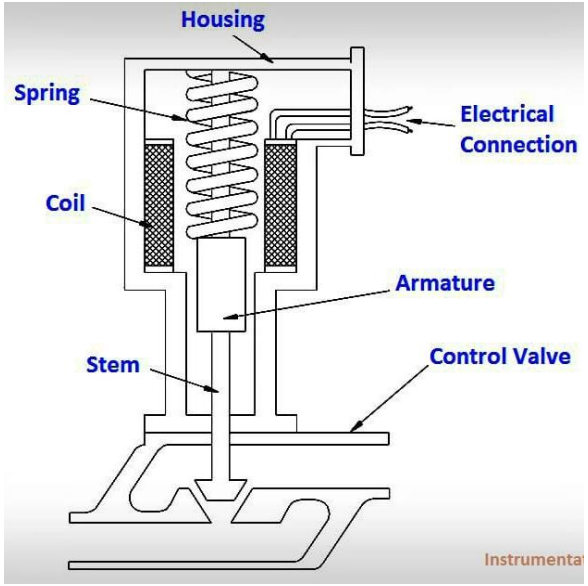
Electronic throttle actuator



- **Principle of operation**: With electronic throttle actuators, there is no direct connection to the accelerator pedal. The driver's desired load is captured by an electronic accelerator pedal (electromotive throttle actuator).
- **Functions** : In the case of petrol engines, speed and power output are regulated by means of fresh air or mixture dosing.
- Diesel engines generally do not need a throttle valve. However, in modern diesel cars, throttling the amount of intake air facilitates precision control for exhaust gas recirculation and stops the engine from shaking when the ignition is switched off.
- The electronic throttle actuator is controlled exclusively using the control signal from the engine management and with the assistance of a servomotor.

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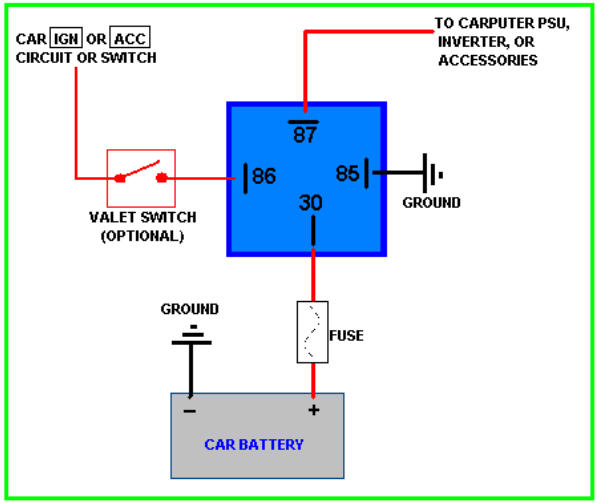
Solenoid valve actuator

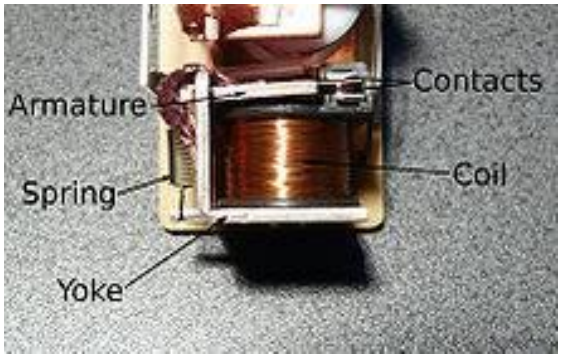


- **Principle of operation**: The coil is connected to an external current supply. The spring rests on the armature to force it downward. The armature moves vertically inside the coil and transmits its motion through the stem to the valve.
- As soon as power is sent through it, a magnetic field gets created which pushes and pulls the stem/plunger in the process of moving the valve.
- **Functions** : Mainly used to regulate the flow of air or a liquid in pneumatic and hydraulic power systems.

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Relay



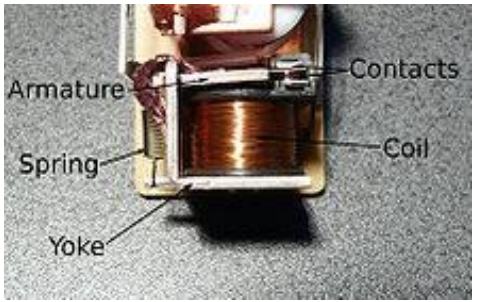


- **Principle of operation**: Relays are electric-powered switch that works with electromagnetism to transform small electric stimuli into large currents.
- This conversion takes place while an electrical current turns on electromagnets to shape or change the current circuits.
- **Functions** : Mainly used to regulate the flow of air or a liquid in pneumatic and hydraulic power systems.

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Components of Automotive Relay

- **Frame –** is a field or heavy-obligation body that includes and helps the numerous components of the relay.
- **Coil –** is a twine wounded around a metallic core. It's the component that reasons an electromagnetic subject
- **Armature –** is a switching component that opens and closes the contacts. There is a connected spring that returns the armature to its authentic role.
- **Contacts –** it's the carrying out of a component that reasons the relay to make (closes) or breaks (opens) a circuit.



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Types of relay

Normally Open (NO) Relays

- Most relays are normally open; this is, the second one, the large circuit is in the off role by the means of default.
- In a normally open relay, power flows via an enter circuit, activating an electromagnet. This generates a magnetic subject that draws a touch to enroll in with the second one. A large circuit, permitting the current to go with the drift. When the supply of power is removed, a spring attracts. The touch far from the second circuit, preventing the go with the drift of energy and turning off the give-up device.

Normally Closed (NC) Relays

- The basics of an NC relay are similar to an NO relay: there are circuits, with the second one being large, and an electromagnet moves a bodily touch among positions.
- But in the case of an NC relay, the default states are reversed. When the primary circuit is activated, the electromagnet attracts the touch far from the second circuit. As such, NC relays hold the bigger circuit via means of default.

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Devise Electronics Pvt. Ltd.

Our Presence: **Pune, Nagpur, and Bengaluru**

Ph No: **+91 - 8767169105**

Email: ***info@deviseelectronics.com***

Website: **www.deviseelectronics.com**

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